Problem Solving using Vectors

Exam code: 9MA0

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Problem solving using vectors

How are vectors used for problem solving?

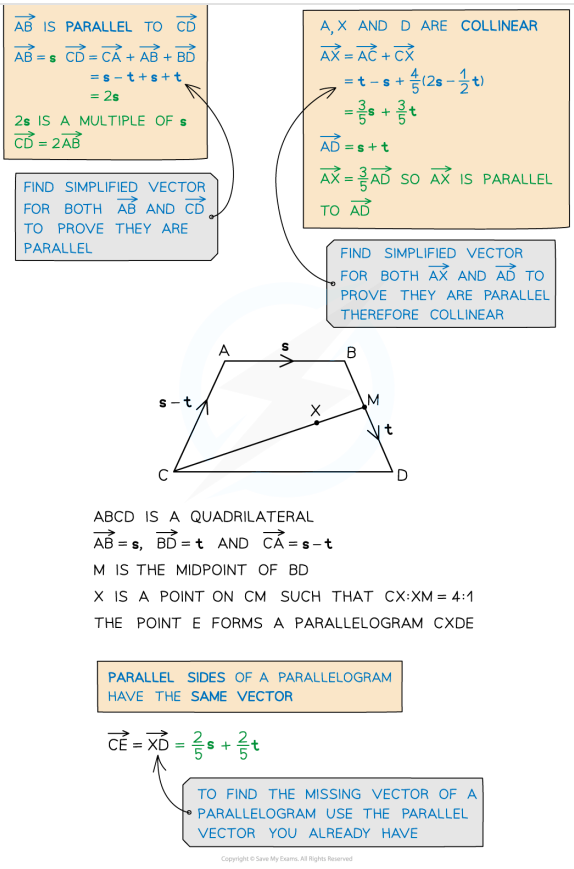
- Vectors can be used to prove two lines are **parallel** (see Vector Addition)
- They can also be used to show points are **collinear** (lie on the same straight line)
- Vectors can be used to find missing vertices of a given shape
- You will need a good understanding of how to divide a line segment into a given ratio

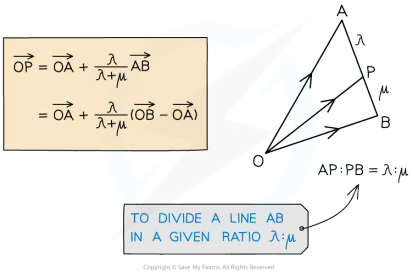
Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

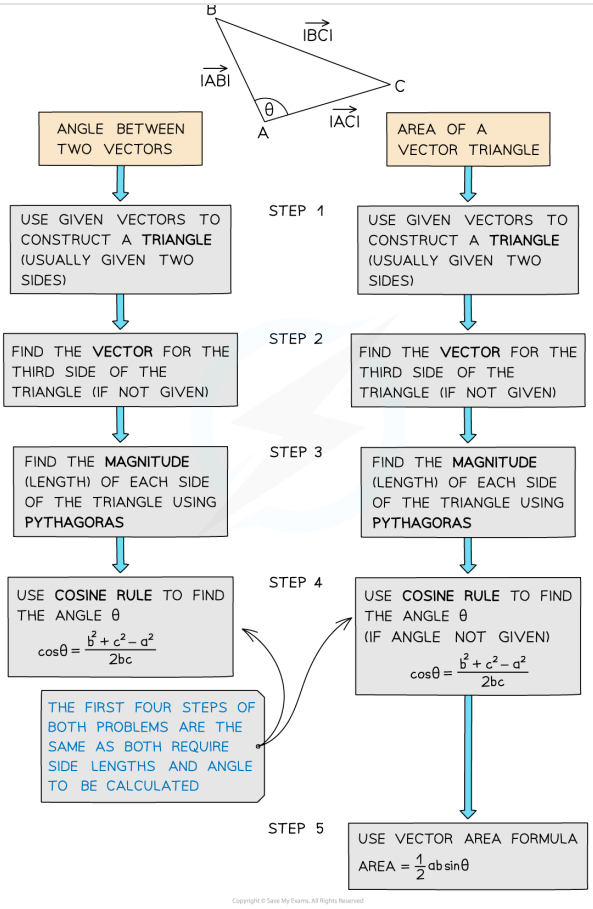
Got it





When do I use trigonometry in vector problems?

- When problem-solving with vectors, trigonometry can help us:
  - convert between **component form** and magnitude/direction form (see Magnitude Direction)
  - find the **angle between two vectors** using **Cosine Rule** (see Non-Right-Angled Triangles)
  - find the **area of a triangle** using a variation of **Area Formula** (see Non-Right-Angled Triangles)



Examiner Tips and Tricks

- Think of vectors like a journey from one place to another – you may have to take a detour eg. A to B might be A to O then O to B.
- Diagrams can help, if there isn't one, draw one. For a given diagram labelling all known vectors and quantities will help.

Worked Example

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